Cryptocurrencies, powered by blockchain technology, generated tremendous excitement over the past few years. Investors and companies dreamt of a future where these digital currencies would power new decentralised use cases and low-latency transactions, with the instantaneous cross-border transfer of money being a much-revered goal. Yet, while 2022 was a tough year for most asset classes globally, it was far worse for cryptocurrencies.

From a peak of close to $3 trillion in November 2021, the aggregate value of all cryptocurrencies has fallen to $800 billion today, implying a loss of over $2 trillion of investor wealth. Among other factors, a series of seemingly fraudulent behaviours—due, in part, to the lack of governance, diligence from investors, and regulations in this space—hurt retail and institutional investors alike. Noteworthy events included:

- The $8 billion FTX scam, which resulted from alleged criminal fraud and money laundering, according to US federal prosecutors;
- The collapse of Terra, which resulted from the disregard of publicly highlighted risks around the currency's algorithmic nature, wiping out approximately $50 billion in market value;
- The alleged nexus between cryptocurrency founders and investors, which resulted in artificially maintained pricing of select coins.

In almost all cases, there was usually either an individual's interest or greed that undid the entire system or extremely poor governance with a complete lack of checks and balances. Notwithstanding the impact of these major setbacks in crypto, what remains true is that the underlying technology—blockchain—is important for our future. People often confuse blockchain with cryptocurrency, but the two are not the same. And the hype and misuse that blockchain has been subjected to in the past few years, in the same way as AI has been overtalked and misused, has stymied full understanding of its true potential and limitations.

To tease out what is meant by blockchain technology, we work through a few common questions:

- What is a blockchain? Simply put, it is a ledger. Yes, the kind that we wrote down stuff on 50 years ago. Except in this case, the ledger is immutable—that is, it can only be written into, and you cannot erase and tamper with it.
- Where does the ledger sit? This ledger is distributed, meaning it is split across many computers with multiple copies of that ledger sitting on each computer across the internet. These computers may be owned by individuals or organisations, and most blockchains have rules around how heterogenous the network of actors must be.
- How is this ledger maintained? There is a protocol, or a set of rules governed by an algorithm, for what information it can store, who can write into it, and how it is updated. These rules are most often programmed by the founders of the blockchain, with that code being open to the public.
- What can you record on the ledger? Pretty much anything: financial transactions, job history, personal identifiers, etc.
- Why use a blockchain to record information? Because the blockchain ledger copies are immutable and maintained across computers, they are thought to be more transparent and trustworthy. And because they are governed by rules, there are perceived efficiencies from the disintermediation that arises by eliminating the need for centralised companies to maintain the ledger.

This barebones description of a blockchain highlights why the world has been excited about them: they offer a distributed way of managing information without the need for a central coordinating entity. That is, blockchains are decentralised. But why is there this fascination for distributed anything? Computer scientists, among others, have long been inspired by the largest and oldest distributed system—nature. There is no single central entity regulating nature and instead it is the most scalable and consistent distributed system we know of, governed by a set of rules and laws. For example, there is no central entity instructing a bee to pollinate a flower, or a flower to turn towards sunlight, a plant

WHILE CRYPTO HAS SUFFERED MAJOR SETBACKS IN 2022, **OTHER APPLICATIONS OF BLOCKCHAIN ARE BEING ACTIVELY EXPLORED** THAT HOLD GREAT PROMISE FOR INDIA AND THE WORLD IN 2023